

NOTE The type test may also be considered valid if it is made on a machine which has minor deviations of rating or other characteristics. These deviations should be subject to agreement.

3.32

routine test

a test to which each individual machine is subjected during or after manufacture to ascertain whether it complies with certain criteria

[IEV 411-53-02]

3.33

runaway speed

the maximum speed attained by the engine/generator set after removal of the full load of the generator if the speed regulator does not function

[IEV 811-17-23]

NOTE For motors, the maximum overspeed at loss of supply is meant that a motor might reach driven by the coupled equipment

4 Duty

4.1 Declaration of duty

It is the responsibility of the purchaser to declare the duty. The purchaser may describe the duty by one of the following:

- a) numerically, where the load does not vary or where it varies in a known manner;
- b) as a time sequence graph of the variable quantities;
- c) by selecting one of the duty types S1 to S10 that is no less onerous than the expected duty.

The duty type shall be designated by the appropriate abbreviation, specified in 4.2, written after the value of the load.

An expression for the cyclic duration factor is given in the relevant duty type figure.

The purchaser normally cannot provide values for the moment of inertia of the motor (J_M) or the relative thermal life expectancy (TL), see Annex A. These values are provided by the manufacturer.

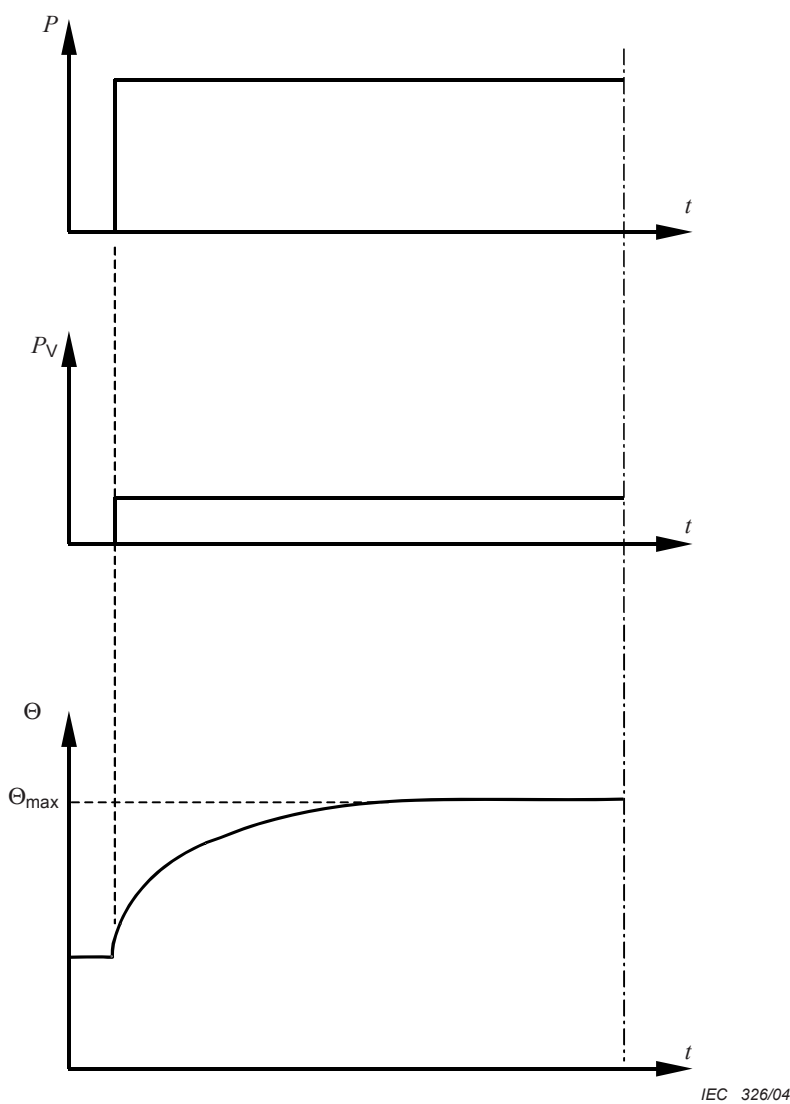
Where the purchaser does not declare a duty, the manufacturer shall assume that duty type S1 (continuous running duty) applies.

4.2 Duty types

4.2.1 Duty type S1 – Continuous running duty

Operation at a constant load maintained for sufficient time to allow the machine to reach thermal equilibrium, see Figure 1.

The appropriate abbreviation is S1.



Key

P	load
P_V	electrical losses
Θ	temperature
$\Theta_{\max}$	maximum temperature attained
t	time

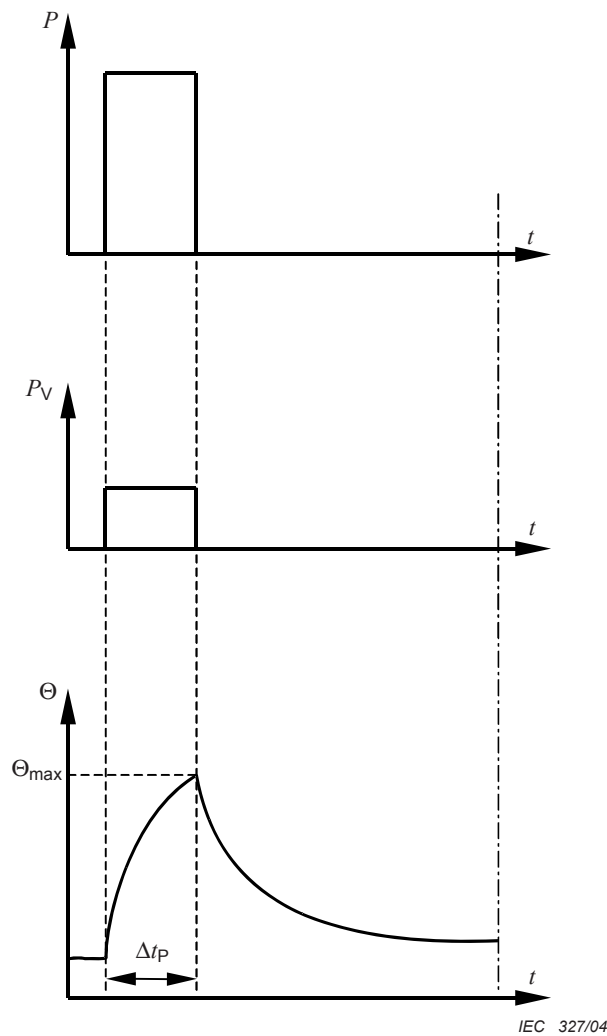
Figure 1 – Continuous running duty – Duty type S1

4.2.2 Duty type S2 – Short-time duty

Operation at constant load for a given time, less than that required to reach thermal equilibrium, followed by a time de-energized and at rest of sufficient duration to re-establish machine temperatures within 2 K of the coolant temperature, see Figure 2.

The appropriate abbreviation is S2, followed by an indication of the duration of the duty,

Example: S2 60 min.



Key

P	load
P_V	electrical losses
Θ	temperature
$\Theta_{\max}$	maximum temperature attained
t	time
Δt_P	operation time at constant load

Figure 2 – Short-time duty – Duty type S2

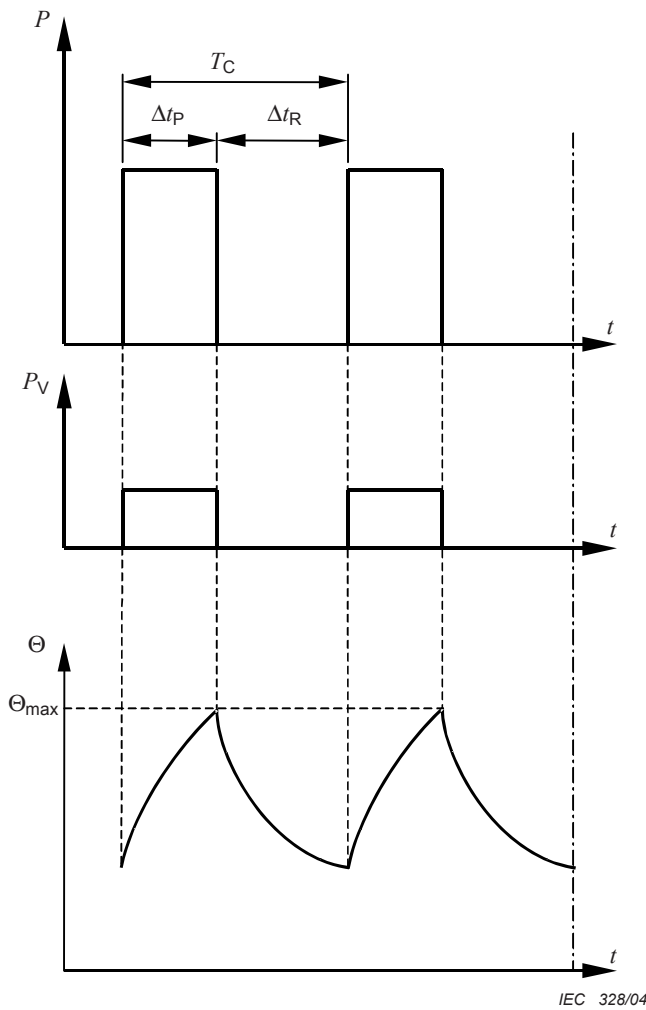
4.2.3 Duty type S3 – Intermittent periodic duty

NOTE Periodic duty implies that thermal equilibrium is not reached during the time on load.

A sequence of identical duty cycles, each including a time of operation at constant load and a time de-energized and at rest, see Figure 3. In this duty, the cycle is such that the starting current does not significantly affect the temperature rise.

The appropriate abbreviation is S3, followed by the cyclic duration factor.

Example: S3 25 %



Key

P	load
P_V	electrical losses
Θ	temperature
$\Theta_{\max}$	maximum temperature attained
t	time
T_C	time of one load cycle
Δt_P	operation time at constant load
Δt_R	time de-energized and at rest
Cyclic duration factor = $\Delta t_P / T_C$	

Figure 3 – Intermittent periodic duty – Duty type S3

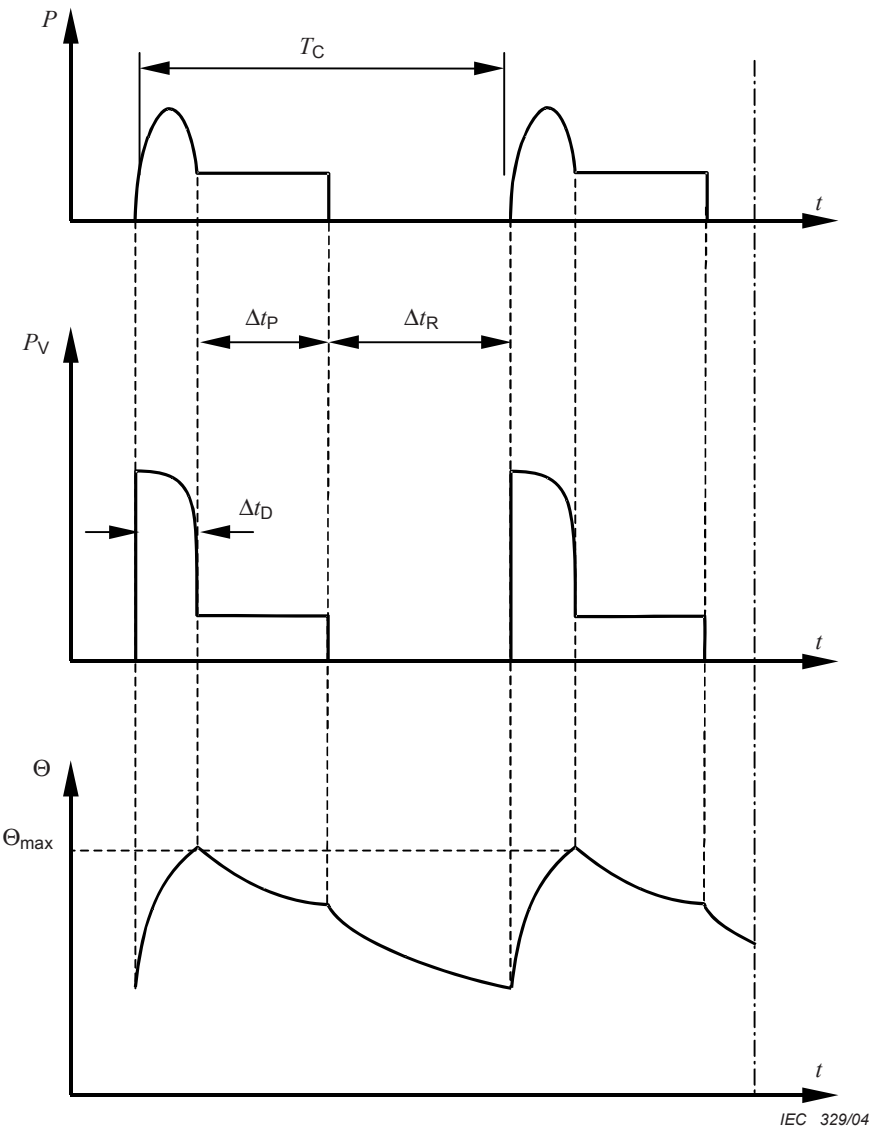
4.2.4 Duty type S4 – Intermittent periodic duty with starting

NOTE Periodic duty implies that thermal equilibrium is not reached during the time on load.

A sequence of identical duty cycles, each cycle including a significant starting time, a time of operation at constant load and a time de-energized and at rest, see Figure 4.

The appropriate abbreviation is S4, followed by the cyclic duration factor, the moment of inertia of the motor (J_M) and the moment of inertia of the load (J_{ext}), both referred to the motor shaft.

Example: S4 25 % $J_M = 0,15 \text{ kg} \times \text{m}^2$ $J_{ext} = 0,7 \text{ kg} \times \text{m}^2$



Key

P	load	t	time
P_V	electrical losses	T_C	time of one load cycle
Θ	temperature	Δt_D	starting/accelerating time
Θ_{max}	maximum temperature attained	Δt_P	operation time at constant load
		Δt_R	time de-energized and at rest

Cyclic duration factor = $(\Delta t_D + \Delta t_P)/T_C$

Figure 4 – Intermittent periodic duty with starting – Duty type S4

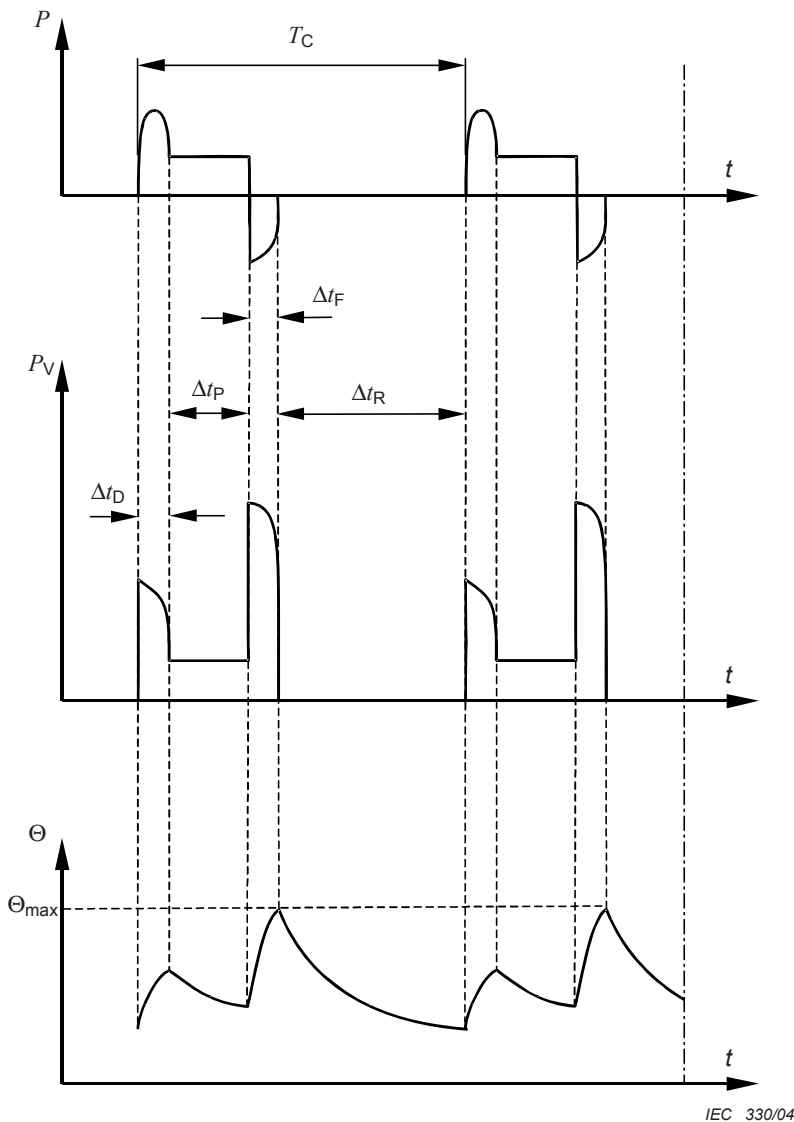
4.2.5 Duty type S5 – Intermittent periodic duty with electric braking

NOTE Periodic duty implies that thermal equilibrium is not reached during the time on load.

A sequence of identical duty cycles, each cycle consisting of a starting time, a time of operation at constant load, a time of electric braking and a time de-energized and at rest, see Figure 5.

The appropriate abbreviation is S5, followed by the cyclic duration factor, the moment of inertia of the motor (J_M) and the moment of inertia of the load (J_{ext}), both referred to the motor shaft.

Example: S5 25 % $J_M = 0,15 \text{ kg} \times \text{m}^2$ $J_{ext} = 0,7 \text{ kg} \times \text{m}^2$



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Key

P	load	T_C	time of one load cycle
P_V	electrical losses	Δt_D	starting/accelerating time
Θ	temperature	Δt_P	operation time at constant load
Θ_{max}	maximum temperature attained	Δt_F	time of electric braking
t	time	Δt_R	time de-energized and at rest

Cyclic duration factor = $(\Delta t_D + \Delta t_P + \Delta t_F)/T_C$

Figure 5 – Intermittent periodic duty with electric braking – Duty type S5

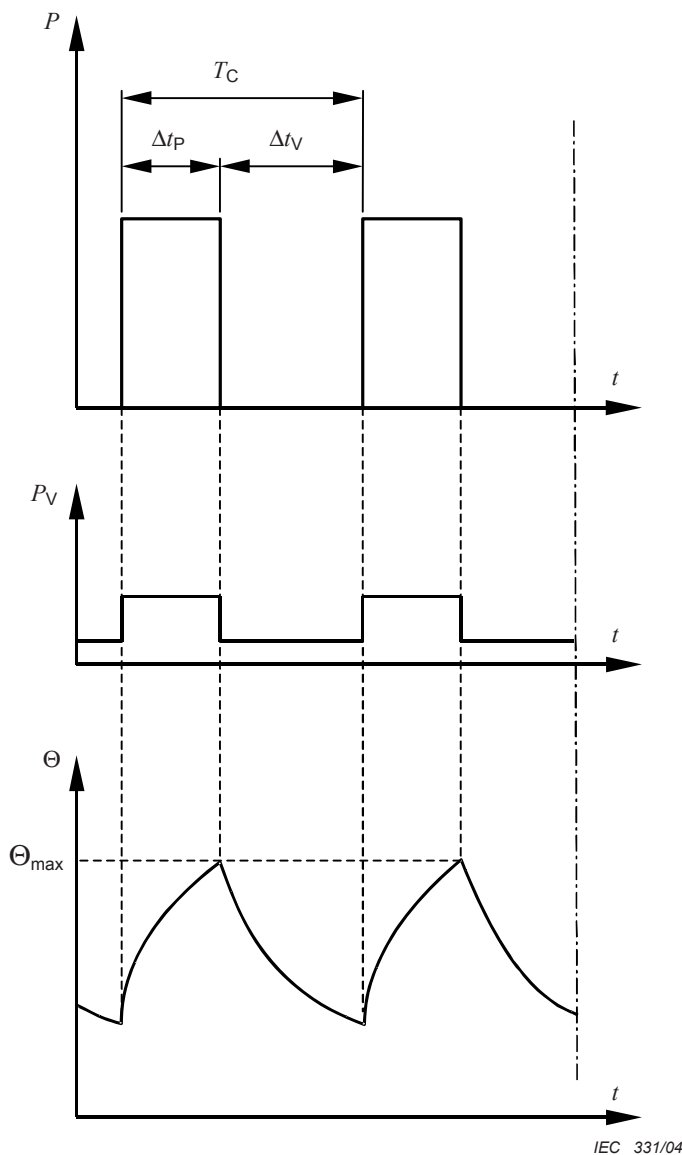
4.2.6 Duty type S6 – Continuous operation periodic duty

NOTE Periodic duty implies that thermal equilibrium is not reached during the time on load.

A sequence of identical duty cycles, each cycle consisting of a time of operation at constant load and a time of operation at no-load. There is no time de-energized and at rest, see Figure 6.

The appropriate abbreviation is S6, followed by the cyclic duration factor.

Example: S6 40 %



Key

P	load	t	time
P_V	electrical losses	T_C	time of one load cycle
Θ	temperature	Δt_P	operation time at constant load
$\Theta_{\max}$	maximum temperature attained	Δt_V	operation time at no-load
Cyclic duration factor = $\Delta t_P/T_C$			

Figure 6 – Continuous operation periodic duty – Duty type S6

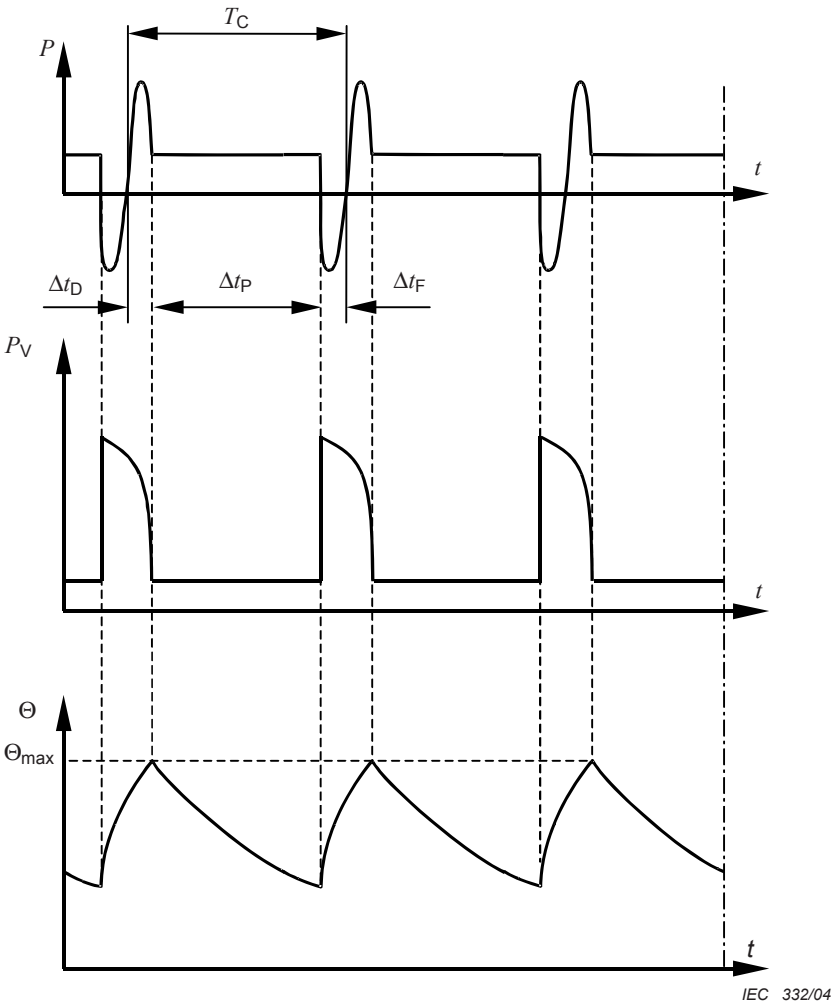
4.2.7 Duty type S7 – Continuous operation periodic duty with electric braking

NOTE Periodic duty implies that thermal equilibrium is not reached during the time on load.

A sequence of identical duty cycles, each cycle consisting of a starting time, a time of operation at constant load and a time of electric braking. There is no time de-energized and at rest, see Figure 7.

The appropriate abbreviation is S7, followed by the moment of inertia of the motor (J_M) and the moment of inertia of the load (J_{ext}), both referred to the motor shaft.

Example: S7 $J_M = 0,4 \text{ kg} \times \text{m}^2$ $J_{ext} = 7,5 \text{ kg} \times \text{m}^2$



Key

P	load	t	time
P_V	electrical losses	T_C	time of one load cycle
Θ	temperature	Δt_D	starting/accelerating time
Θ_{max}	maximum temperature attained	Δt_P	operation time at constant load
Cyclic duration factor = 1		Δt_F	time of electric braking

Figure 7 – Continuous operation periodic duty with electric braking – Duty type S7

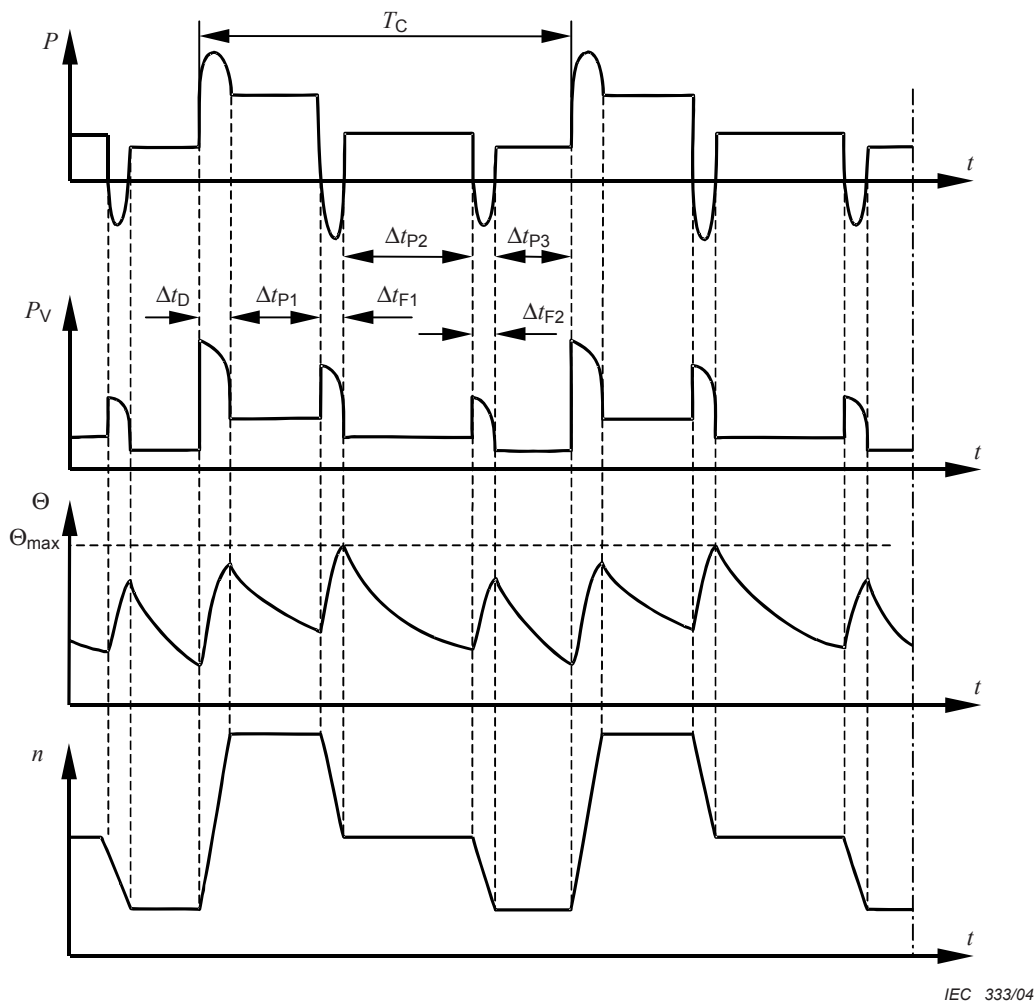
4.2.8 Duty type S8 – Continuous operation periodic duty with related load/speed changes

NOTE Periodic duty implies that thermal equilibrium is not reached during the time on load.

A sequence of identical duty cycles, each cycle consisting of a time of operation at constant load corresponding to a predetermined speed of rotation, followed by one or more times of operation at other constant loads corresponding to different speeds of rotation (carried out, for example, by means of a change in the number of poles in the case of induction motors). There is no time de-energized and at rest (see Figure 8).

The appropriate abbreviation is S8, followed by the moment of inertia of the motor (J_M) and the moment of inertia of the load (J_{ext}), both referred to the motor shaft, together with the load, speed and cyclic duration factor for each speed condition.

Example:	S8 $J_M = 0,5 \text{ kg} \times \text{m}^2$	$J_{ext} = 6 \text{ kg} \times \text{m}^2$	16 kW	740 min ⁻¹	30 %
			40 kW	1 460 min ⁻¹	30 %
			25 kW	980 min ⁻¹	40 %



Key

P	load	t	time
P_V	electrical losses	T_C	time of one load cycle
Θ	temperature	Δt_D	starting/accelerating time
Θ_{max}	maximum temperature attained	Δt_P	operation time at constant load (P1, P2, P3)
n	speed	Δt_F	time of electric braking (F1, F2)

Cyclic duration factor = $(\Delta t_D + \Delta t_{P1})/T_C$; $(\Delta t_{F1} + \Delta t_{P2})/T_C$; $(\Delta t_{F2} + \Delta t_{P3})/T_C$

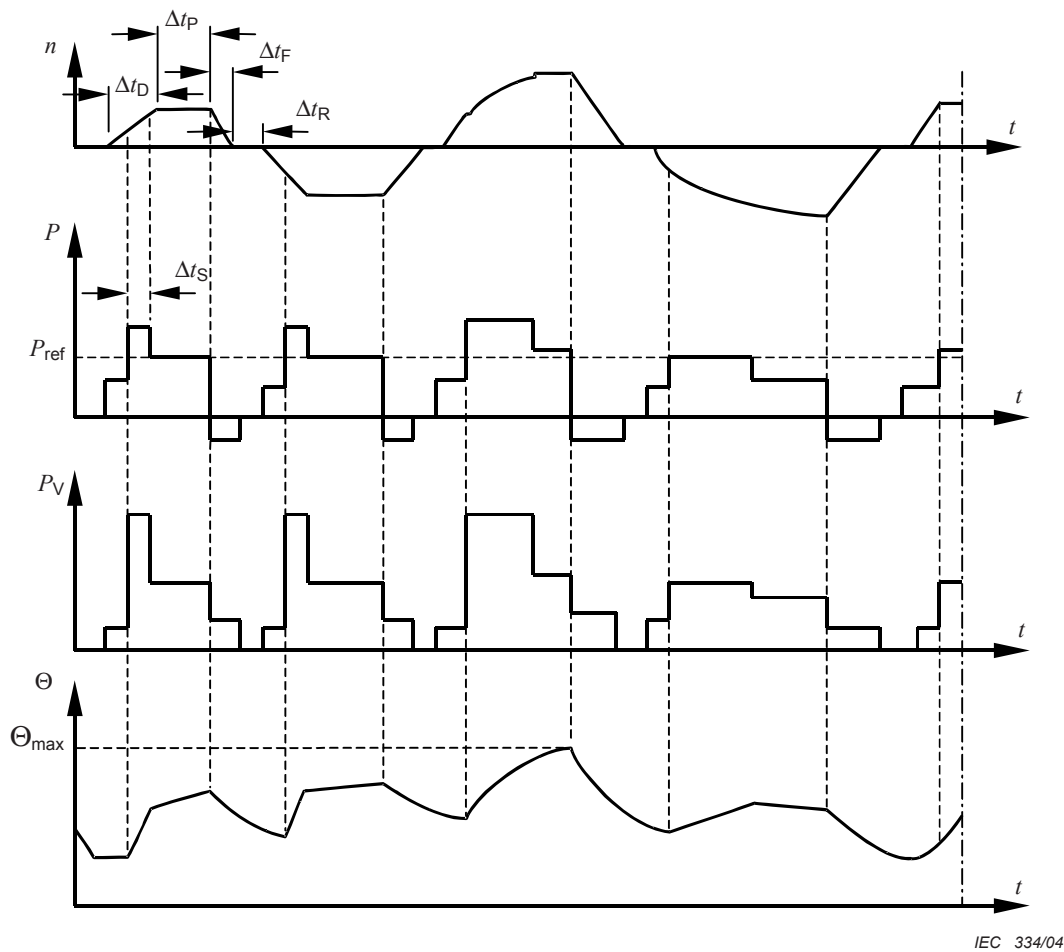
Figure 8 – Continuous operation periodic duty with related load/speed changes – Duty type S8

4.2.9 Duty type S9 – Duty with non-periodic load and speed variations

A duty in which generally load and speed vary non-periodically within the permissible operating range. This duty includes frequently applied overloads that may greatly exceed the reference load (see Figure 9).

The appropriate abbreviation is S9.

For this duty type, a constant load appropriately selected and based on duty type S1 is taken as the reference value (" P_{ref} " in Figure 9) for the overload concept.



Key

P	load	t	time
P_{ref}	reference load	Δt_D	starting/accelerating time
P_V	electrical losses	Δt_P	operation time at constant load
Θ	temperature	Δt_F	time of electric braking
Θ_{max}	maximum temperature attained	Δt_R	time de-energized and at rest
n	speed	Δt_S	time under overload

Figure 9 – Duty with non-periodic load and speed variations – Duty type S9

4.2.10 Duty type S10 – Duty with discrete constant loads and speeds

A duty consisting of a specific number of discrete values of load (or equivalent loading) and if applicable, speed, each load/speed combination being maintained for sufficient time to allow the machine to reach thermal equilibrium, see Figure 10. The minimum load within a duty cycle may have the value zero (no-load or de-energized and at rest).

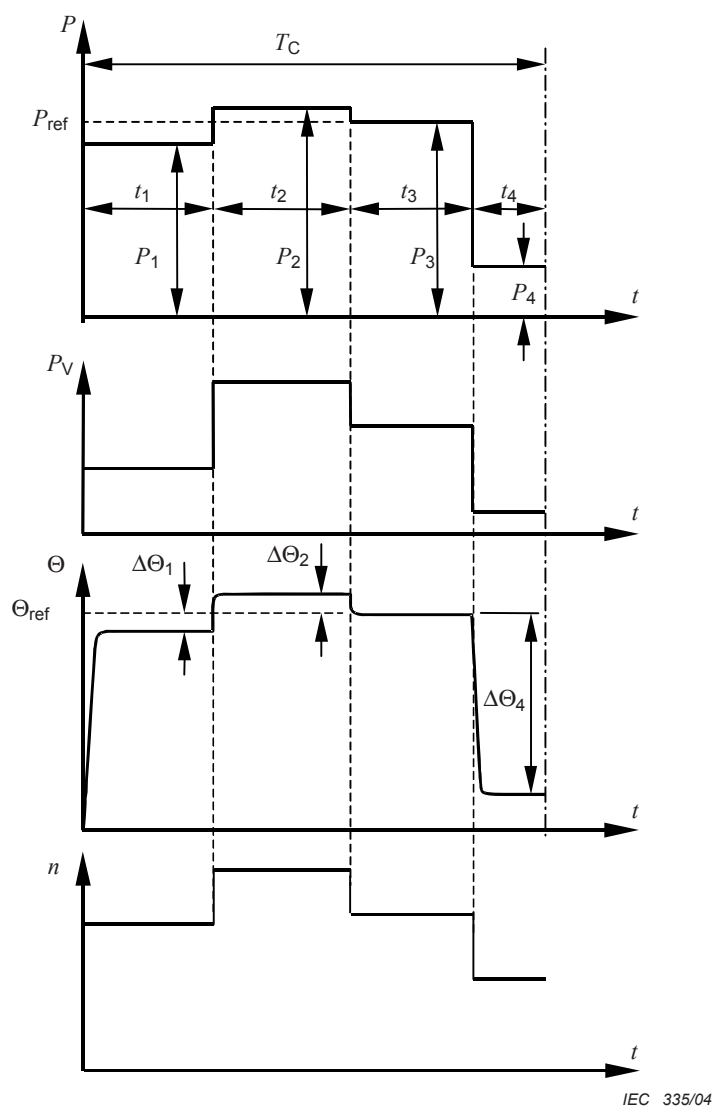
The appropriate abbreviation is S10, followed by the per unit quantities $p/\Delta t$ for the respective load and its duration and the per unit quantity TL for the relative thermal life expectancy of the insulation system. The reference value for the thermal life expectancy is the thermal life expectancy at rating for continuous running duty and permissible limits of temperature rise based on duty type S1. For a time de-energized and at rest, the load shall be indicated by the letter r .

Example: S10 $p/\Delta t = 1,1/0,4; 1/0,3; 0,9/0,2; r/0,1$ $TL = 0,6$

The value of TL should be rounded off to the nearest multiple of 0,05. Advice concerning the significance of this parameter and the derivation of its value is given in Annex A.

For this duty type a constant load appropriately selected and based on duty type S1 shall be taken as the reference value (P_{ref} in Figure 10) for the discrete loads.

NOTE The discrete values of load will usually be equivalent loading based on integration over a period of time. It is not necessary that each load cycle be exactly the same, only that each load within a cycle be maintained for sufficient time for thermal equilibrium to be reached, and that each load cycle be capable of being integrated to give the same relative thermal life expectancy.



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Key

P	load	t	time
P_i	constant load within a load cycle	t_i	time of a constant load within a cycle
P_{ref}	reference load based on duty type S1	T_C	time of one load cycle
P_V	electrical losses	$\Delta\Theta_i$	difference between the temperature rise of the winding at each of the various loads within one cycle and the temperature rise based on duty cycle S1 with reference load
Θ	temperature	n	speed
Θ_{ref}	temperature at reference load based on duty type S1		

Figure 10 – Duty with discrete constant loads – Duty type S10